# Hypothesis Testing

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational] • Target: The logic of hypothesis testing**

State the null and alternative hypotheses for testing whether a new fertilizer increases the mean crop yield $\mu$, given a historical mean $\mu_0 = 50$ kg. Classify the test as one-tailed or two-tailed.

$H_0 : \mu = 50$. $H_a : \mu > 50$. This is a one-tailed (right-tailed) test, since the alternative specifies a direction.

**2** **[Foundational] • Target: The logic of hypothesis testing**

Explain the difference between a one-tailed and a two-tailed hypothesis test, giving an example scenario for each.

A one-tailed test is used when the alternative hypothesis specifies a direction (e.g. testing whether a new drug \emph{increases} recovery rate). A two-tailed test is used when the alternative simply states the parameter differs from the claimed value, without specifying direction (e.g. testing whether a machine's mean output has changed at all, up or down).

**3** **[Foundational] • Target: The logic of hypothesis testing**

A test has $H_0 : \mu = 75$ vs $H_a : \mu \neq 75$, and the sample data leads to rejecting $H_0$ at the 5% significance level. State, without further calculation, whether this data would also lead to rejection at the 1% level, justifying your answer.

Not necessarily — a stricter significance level (1%) requires stronger evidence (a smaller $p$-value) to reject $H_0$. If the $p$-value is very small (well below $0.01$), rejection would still occur at 1%; but if the $p$-value lies between $0.01$ and $0.05$, $H_0$ would be rejected at 5% but not at 1%.

**4** **[Foundational] • Target: Hypothesis test for a population mean (known σ)**

A machine fills bags with a claimed mean of $500$g, with known population standard deviation $\sigma = 15$g. A sample of $n = 36$ bags has mean $\bar{x} = 494$g. Test at the 5% level whether the mean is less than 500g, using $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ and critical value $-1.645$.

$z = \frac{494 - 500}{15/\sqrt{36}} = \frac{-6}{2.5} = -2.4$. Since $-2.4 < -1.645$, we reject $H_0$. There is significant evidence at the 5% level that the mean fill weight is less than $500$g.

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**5** [Foundational] • Target: Hypothesis test for a population mean (known σ)

A population has known standard deviation $\sigma = 8$. A sample of $n = 64$ has mean $\bar{x} = 103$, against a claimed mean $\mu_0 = 100$. Test at the 1% level whether the mean exceeds 100, using critical value $z^* = 2.326$.

$z = \frac{103 - 100}{8/\sqrt{64}} = \frac{3}{1} = 3$. Since $3 > 2.326$, we reject $H_0$. There is significant evidence at the 1% level that the mean exceeds $100$.

**6** [Foundational] • Target: Two-tailed tests for a population mean

A standardized test historically has mean $\mu_0 = 100$ with $\sigma = 15$. A sample of $n = 25$ students taking a new prep course has mean $\bar{x} = 107$. Test at the 5% level (two-tailed) whether the course changes the mean score, using critical values $\pm 1.96$.

$z = \frac{107 - 100}{15/\sqrt{25}} = \frac{7}{3} \approx 2.333$. Since $|2.333| > 1.96$, we reject $H_0$. There is significant evidence at the 5% level that the prep course changes the mean score.

**7** [Foundational] • Target: Two-tailed tests for a population mean

Using $z \approx 2.33$ from the previous problem, find the two-tailed $p$-value, given $\Phi(2.33) \approx 0.9901$, and confirm the conclusion.

$p-\text{value} = 2(1 - \Phi(2.33)) = 2(1 - 0.9901) = 2(0.0099) = 0.0198$. Since $0.0198 < 0.05$, we reject $H_0$, confirming the earlier conclusion.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Two-tailed tests for a population mean

A sample of $n = 50$ has mean $\bar{x} = 48.2$, against a claimed mean $\mu_0 = 50$ with $\sigma = 6$. Test at the 5% level (two-tailed) whether the mean differs from 50, using critical values $\pm 1.96$.

$z = \frac{48.2 - 50}{6/\sqrt{50}} = \frac{-1.8}{0.8485} \approx -2.121$. Since $|-2.121| > 1.96$, we reject $H_0$. There is significant evidence at the 5% level that the mean differs from $50$.

**9** [Intermediate] • Target: Hypothesis test for a Poisson mean

A website historically crashes at a rate of $\lambda_0 = 4$ per month. After a software update, 1 crash is observed in a randomly selected month. Test at the 5% level whether the crash rate has decreased, using $P(X \le 1) \approx 0.0916$ for $X \sim \text{Po}(4)$.

$H_0 : \lambda = 4$. $H_a : \lambda < 4$. $p-\text{value} = P(X \le 1) = 0.0916$. Since $0.0916 > 0.05$, we do not reject $H_0$. There is insufficient evidence at the 5% level that the crash rate has decreased.

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**10** [Intermediate] • Target: Hypothesis test for a Poisson mean

Explain why an exact Poisson probability, rather than a normal approximation, is used to test a hypothesis about a Poisson mean when $\lambda_0$ is small.

When $\lambda_0$ is small, the Poisson distribution is noticeably skewed and a normal approximation would be unreliable, so exact Poisson probabilities give a more accurate $p$-value.

**11** [Intermediate] • Target: Further Poisson mean tests

A call center historically receives calls at $\lambda_0 = 6$ per hour. During a promotional event, 12 calls are received in one hour. Test at the 5% level whether the call rate has increased, using $P(X \geq 12) \approx 0.0201$ for $X \sim \text{Po}(6)$.

$H_0 : \lambda = 6$. $H_a : \lambda > 6$. $p\text{-value} = P(X \geq 12) = 0.0201$. Since $0.0201 < 0.05$, we reject $H_0$. There is significant evidence at the 5% level that the call rate has increased during the promotional event.

**12** [Intermediate] • Target: Further Poisson mean tests

A factory's defect rate is historically $\lambda_0 = 3$ per shift. During a trial of new equipment, 8 defects are observed in one shift. Test at the 5% level whether the defect rate has increased, using $P(X \geq 8) \approx 0.0119$ for $X \sim \text{Po}(3)$.

$H_0 : \lambda = 3$. $H_a : \lambda > 3$. $p\text{-value} = P(X \geq 8) = 0.0119$. Since $0.0119 < 0.05$, we reject $H_0$. There is significant evidence at the 5% level that the defect rate has increased.

**13** [Intermediate] • Target: Hypothesis test for a binomial proportion

A company claims $40\%$ of its customers use its mobile app ($p_0 = 0.4$). In a sample of $n = 15$ customers, 10 report using the app. Test at the 5% level whether the true proportion is greater than 40%, using $P(X \geq 10) \approx 0.0338$ for $X \sim B(15, 0.4)$.

$H_0 : p = 0.4$. $H_a : p > 0.4$. $p\text{-value} = P(X \geq 10) = 0.0338$. Since $0.0338 < 0.05$, we reject $H_0$. There is significant evidence at the 5% level that more than 40% of customers use the app.

**14** [Intermediate] • Target: Hypothesis test for a binomial proportion

Explain why the binomial hypothesis test uses the exact binomial distribution rather than a normal approximation when $n$ is small.

For small $n$, the binomial distribution can be quite skewed (especially if $p_0$ is far from $0.5$), so a normal approximation would be unreliable; exact binomial probabilities give a more accurate $p$-value.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 2.01 Exponentials & Logarithms

A coin is tested for fairness ($p_0 = 0.5$). In 20 tosses, 6 heads are observed. Test at the 5% level (two-tailed) whether the coin is biased, using $P(X \leq 6) \approx 0.0577$ for $X \sim B(20, 0.5)$ (doubled for a two-tailed test).

$H_0 : p = 0.5$. $H_a : p \neq 0.5$. Two-tailed $p\text{−value} = 2 \times P(X \leq 6) = 2(0.0577) = 0.1154$. Since $0.1154 > 0.05$, we do not reject $H_0$. There is insufficient evidence at the 5% level that the coin is biased.

**16** [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 3.01 Advanced Trigonometry

A politician claims $60\%$ of voters support a new policy ($p_0 = 0.6$). In a sample of 12 voters, only 3 support it. Test at the 5% level whether support is less than 60%, using $P(X \leq 3) \approx 0.0153$ for $X \sim B(12, 0.6)$.

$H_0 : p = 0.6$. $H_a : p < 0.6$. $p\text{−value} = P(X \leq 3) = 0.0153$. Since $0.0153 < 0.05$, we reject $H_0$. There is significant evidence at the 5% level that support is less than 60%.

**17** [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 4.01 Differentiation

In the website crash-rate test, describe what a Type I error would mean in context, and what a Type II error would mean.

A Type I error would mean concluding the crash rate has decreased when it actually hasn't (rejecting a true $H_0$) — wrongly crediting the software update. A Type II error would mean failing to detect a real decrease in the crash rate (failing to reject a false $H_0$) — missing genuine improvement from the update.

**18** [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 5.01 Integration

Explain what is meant by the significance level of a hypothesis test, and state the probability of a Type I error in a test conducted at the 5% significance level.

The significance level $\alpha$ is the probability of rejecting a true null hypothesis (a Type I error), chosen in advance by the person conducting the test. At the 5% significance level, $\alpha = 0.05$, so there is a 5% probability of a Type I error.

**19** [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 6.01 Sequences & Series

A researcher conducts a test at $\alpha = 0.10$ instead of the more common $\alpha = 0.05$. Explain how this affects the probability of Type I and Type II errors, compared to using $\alpha = 0.05$.

Using a larger $\alpha = 0.10$ increases the probability of a Type I error (the rejection region is larger, so a true $H_0$ is more likely to be rejected by chance) but decreases the probability of a Type II error (it becomes easier to reject $H_0$, so a false $H_0$ is more likely to be correctly rejected) — there is a direct tradeoff between the two error types as $\alpha$ changes.

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**20** [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 7.01 Vectors

This question has four parts. A factory's current process produces components with mean breaking strength $\mu_0 = 250$ N and known $\sigma = 20$ N. A new process gives a sample of $n = 40$ components with mean $\bar{x} = 257$ N. (a) State $H_0$ and $H_a$ for testing whether the new process increases mean breaking strength (one-tailed). (b) Compute the test statistic $z$. (c) Using critical value $z^* = 1.645$ (5% level), state the conclusion. (d) Quality control also monitors defects; the historical rate is $\lambda_0 = 5$ per batch, but a batch from the new process has only 1 defect. Using $P(X \le 1) \approx 0.0404$ for $X \sim \text{Po}(5)$, test at the 5% level whether the new process has reduced the defect rate.

(a) $H_0 : \mu = 250$. $H_a : \mu > 250$. (b) $z = \frac{257 - 250}{20/\sqrt{40}} = \frac{7}{3.1623} \approx 2.214$. (c) Since $2.214 > 1.645$, we reject $H_0$ : significant evidence at the 5% level that the new process increases mean breaking strength. (d) $H_0 : \lambda = 5$, $H_a : \lambda < 5$. $p\text{-value} = P(X \le 1) = 0.0404 < 0.05$, so we reject $H_0$: significant evidence at the 5% level that the new process has reduced the defect rate.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Null & Alternative Hypotheses, Type I & Type II Errors

Define a Type I error in hypothesis testing.

**a.** **Rejecting $H_0$ when $H_0$ is actually true ✓**

**b.** Failing to reject $H_0$ when $H_0$ is false

**c.** Setting the significance level too high

**d.** Accepting $H_1$ when it is false

A Type I error occurs when a true null hypothesis is incorrectly rejected. The probability of a Type I error equals the significance level $\alpha$.

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**22** [Foundational] • Target: Null & Alternative Hypotheses, Type I & Type II Errors

Define a Type II error in hypothesis testing.

**a. Failing to reject $H_0$ when $H_0$ is false** ✓

**b.** Rejecting $H_0$ when $H_0$ is true

**c.** Computing an incorrect test statistic

**d.** Rejecting $H_1$ when $H_0$ is true

A Type II error (probability $\beta$) occurs when the test fails to reject a false null hypothesis.

**23** [Foundational] • Target: Null & Alternative Hypotheses, Type I & Type II Errors

A machine produces metal pins with mean length 5.0 cm. A technician suspects the machine is out of adjustment and producing pins of different mean length. What are the appropriate hypotheses?

**a.** $H_0 : \mu = 5.0$, $H_1 : \mu \neq 5.0$ ✓

**b.** $H_0 : \mu = 5.0, H_1 : \mu > 5.0$

**c.** $H_0 : \mu \neq 5.0, H_1 : \mu = 5.0$

**d.** $H_0 : \mu = 5.0, H_1 : \mu < 5.0$

A non-directional suspicion ('different mean length') requires a two-tailed alternative hypothesis $H_1 : \mu \neq 5.0$.

**24** [Foundational] • Target: Testing Population Mean with Normal Distribution

A test of $H_0 : \mu = 80$ against $H_1 : \mu > 80$ at the 5% level is conducted. A sample of $n = 36$ from $N(\mu, 144)$ gives $\bar{x} = 83.5$. What is the value of the test statistic $z$?

**a.** $z = 1.75$ ✓

**b.** $z = 1.645$

**c.** $z = 2.00$

**d.** $z = 0.875$

$\sigma = 12$, SE $= 12/\sqrt{36} = 2.0$. Test statistic $z = \frac{83.5 - 80}{2.0} = \frac{3.5}{2.0} = 1.75$.

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**25** [Foundational] • Target: Testing Population Mean with Normal Distribution

For the test in the previous question ($z = 1.75$), what is the test conclusion at the $5\%$ significance level (critical value $1.645$)?

**a. Reject $H_0$; there is sufficient evidence that the mean is greater than 80 ✓**

**b.** Fail to reject $H_0$; insufficient evidence

**c.** Accept $H_0$; the difference is due to chance

**d.** Test is inconclusive

Since $z = 1.75 > 1.645$, the test statistic falls in the critical region. Reject $H_0$ at the 5% level.

**26** [Foundational] • Target: Testing Population Mean with Normal Distribution

What is the $p$-value for the test statistic $z = 1.75$ in a one-tailed test?

**a.** $0.0401$ ✓

**b.** $0.0802$

**c.** $0.9599$

**d.** $0.0250$

$p-\text{value} = 1 - \Phi(1.75) = 1 - 0.9599 = 0.0401$.

**27** [Foundational] • Target: Testing Parameters with Poisson & Binomial Models

A company claims its weekly defect rate is $\lambda = 2$. In a given week, 6 defects are observed. Test $H_0 : \lambda = 2$ against $H_1 : \lambda > 2$ by calculating $P(X \geq 6)$ under $H_0$.

**a.** $0.0166$ ✓

**b.** $0.0527$

**c.** $0.0345$

**d.** $0.0045$

For Po(2): $P(X \leq 5) = e^{-2}(1 + 2 + 2 + 4/3 + 2/3 + 4/15) = e^{-2}(109/15) = 0.135335 \times 7.2667 \approx 0.9834$. $P(X \geq 6) = 1 - 0.9834 = 0.0166$.

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**28** [Foundational] • Target: Testing Parameters with Poisson & Binomial Models

At the 5% significance level, what is the conclusion for the defect test in the previous question ($p = 0.0166$)?

**a.** **Reject $H_0$; the defect rate has significantly increased** ✓

**b.** Fail to reject $H_0$

**c.** Accept $H_0$

**d.** Change the significance level to 1%

Since $p = 0.0166 < 0.05$, reject $H_0$ at the 5% level.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Testing Parameters with Poisson & Binomial Models

A seed packet claims a 90% germination rate. A gardener tests 20 seeds and finds 15 germinate. To test $H_0 : p = 0.9$ against $H_1 : p < 0.9$, which probability must be computed?

**a.** $P(X \leq 15)$ **where** $X \sim B(20, 0.9)$ ✓

**b.** $P(X = 15)$ where $X \sim B(20, 0.9)$

**c.** $P(X \geq 15)$ where $X \sim B(20, 0.9)$

**d.** $P(X \leq 15)$ where $X \sim N(18, 1.8)$

For a lower-tail test ($H_1 : p < 0.9$), the $p$-value is the probability of observing an outcome as extreme as or more extreme than the observed value: $P(X \leq 15)$.

**30** [Intermediate] • Target: Critical Regions & Power of a Test

For a test of $H_0 : \mu = 50$ vs $H_1 : \mu < 50$ using $n = 25, \sigma = 10$, find the critical value for $\bar{x}$ at the 5% level ($z_c = -1.645$).

**a.** 46.71 ✓

**b.** 45.00

**c.** 47.50

**d.** 46.00

Critical value $= \mu_0 - z_c \frac{\sigma}{\sqrt{n}} = 50 - 1.645\left(\frac{10}{5}\right) = 50 - 1.645(2) = 50 - 3.29 = 46.71$.

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**31** [Intermediate] • Target: Critical Regions & Power of a Test

In the previous test (critical region $\bar{X} < 46.71$), what is the probability of a Type II error if the true mean is $\mu = 45$?

**a.** 0.1964 ✓

**b.** 0.8036

**c.** 0.0500

**d.** 0.3085

Type II error $\beta = P(\bar{X} \geq 46.71 \mid \mu = 45)$. Under $\mu = 45$: $z = \frac{46.71 - 45}{2.0} = 0.855$.
$\beta = 1 - \Phi(0.855) = 1 - 0.8037 = 0.1963 \approx 0.1964$.

**32** [Intermediate] • Target: Critical Regions & Power of a Test

What is the statistical power of the test in the previous question (where $\beta = 0.1964$)?

**a.** 0.8036 ✓

**b.** 0.1964

**c.** 0.9500

**d.** 0.0500

Power of a test is defined as $1 - \beta = 1 - 0.1964 = 0.8036$ (or 80.36%).

**33** [Intermediate] • Target: Critical Regions & Power of a Test

Formulate the null and alternative hypotheses for the following scenario: "A factory claims the mean weight of a product is 250g. An inspector wants to test if it has changed (either direction)."

**a.** $H_0 : \mu = 500,\ H_1 : \mu < 500$

**b.** $H_0 : \mu = 3,\ H_1 : \mu > 3$

**c.** $H_0 : \mu = 10,\ H_1 : \mu < 10$

**d.** $H_0 : \mu = 250,\ H_1 : \mu e 250$ ✓

Applying Hypothesis Testing principles, $H_0 : \mu = 250,\ H_1 : \mu e 250$. Therefore, the verified correct option is $H_0 : \mu = 250,\ H_1 : \mu e 250$.

**34** [Intermediate] • Target: The logic of hypothesis testing

Formulate the null and alternative hypotheses for the following scenario: "A school claims the mean test score is 70. A teacher believes it is actually higher."

**a.** $H_0 : \mu = 10,\ H_1 : \mu < 10$

**b.** $H_0 : \mu = 70,\ H_1 : \mu > 70$ ✓

**c.** $H_0 : \mu = 500,\ H_1 : \mu < 500$

**d.** $H_0 : \mu = 3,\ H_1 : \mu > 3$

Applying Hypothesis Testing principles, $H_0 : \mu = 70,\ H_1 : \mu > 70$. Therefore, the verified correct option is $H_0 : \mu = 70,\ H_1 : \mu > 70$.

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**35** [Intermediate] • Target: Hypothesis test for a population mean (known σ)

For a hypothesis test at the 1% (two-tail) significance level, find the critical z-value.

**a.** 1.645

**b.** 2.326

**c.** **2.576** ✓

**d.** 1.96

The critical value for a 1% (two-tail) test is $z = 2.576$.

**36** [Intermediate] • Target: Two-tailed tests for a population mean

For a hypothesis test at the 5% (two-tail) significance level, find the critical z-value.

**a.** 2.576

**b.** 1.645

**c.** **1.96** ✓

**d.** 1.282

The critical value for a 5% (two-tail) test is $z = 1.96$.

**37** [Intermediate] • Target: Hypothesis test for a Poisson mean

Formulate the null and alternative hypotheses for the following scenario: "A manufacturer claims the mean lifetime of a battery is 500 hours. A researcher suspects it is less than this."

**a.** $H_0 : \mu = 500,\ H_1 : \mu < 500$ ✓

**b.** $H_0 : \mu = 70,\ H_1 : \mu > 70$

**c.** $H_0 : \mu = 3,\ H_1 : \mu > 3$

**d.** $H_0 : \mu = 250,\ H_1 : \mu e 250$

Applying Hypothesis Testing principles, $H_0 : \mu = 500,\ H_1 : \mu < 500$. Therefore, the verified correct option is $H_0 : \mu = 500,\ H_1 : \mu < 500$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Further Poisson mean tests • Cross-Lesson: 5.01 Integration

For a hypothesis test at the 10% (one-tail) significance level, find the critical z-value.

**a.** **1.282** ✓

**b.** 2.576

**c.** 2.326

**d.** 1.645

The critical value for a 10% (one-tail) test is $z = 1.282$.

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**39** [Challenging] • Target: Hypothesis test for a binomial proportion • Cross-Lesson: 6.01 Sequences & Series

For a hypothesis test at the 2.5% (one-tail) significance level, find the critical z-value.

**a.** 2.326

**b.** **1.96** ✓

**c.** 1.645

**d.** 1.282

The critical value for a 2.5% (one-tail) test is $z = 1.96$.

**40** [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 7.01 Vectors

For a hypothesis test at the 5% (one-tail) significance level, find the critical z-value.

**a.** 1.96

**b.** **1.645** ✓

**c.** 1.282

**d.** 2.576

The critical value for a 5% (one-tail) test is $z = 1.645$.

**41** [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A hypothesis test is carried out at the 2.5% significance level. Find the probability of a Type I error.

**a.** **2.5\%** ✓

**b.** 50\%

**c.** 0\%

**d.** 97.5\%

By definition, $P(\text{Type I error}) = \alpha = 2.5\%$.

**42** [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 9.01 Complex Numbers

For a hypothesis test at the 1% (one-tail) significance level, find the critical z-value.

**a.** 2.576

**b.** 1.282

**c.** 1.96

**d.** **2.326** ✓

The critical value for a 1% (one-tail) test is $z = 2.326$.

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**43** [Challenging] • Target: Null & Alternative Hypotheses, Type I & Type II Errors • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.

**a.** 90\%

**b.** **10\% ✓**

**c.** 50\%

**d.** 0\%

By definition, $P(\text{Type I error}) = \alpha = 10\%$.

**44** [Challenging] • Target: Testing Population Mean with Normal Distribution • Cross-Lesson: 10.01 Numerical Methods

A hypothesis test is carried out at the 2% significance level. Find the probability of a Type I error.

**a.** **2\% ✓**

**b.** 0\%

**c.** 98\%

**d.** 50\%

By definition, $P(\text{Type I error}) = \alpha = 2\%$.

**45** [Challenging] • Target: Testing Parameters with Poisson & Binomial Models • Cross-Lesson: 11.01 Differential Equations

A hypothesis test is carried out at the 1% significance level. Find the probability of a Type I error.

**a.** **1\% ✓**

**b.** 50\%

**c.** 99\%

**d.** 0\%

By definition, $P(\text{Type I error}) = \alpha = 1\%$.

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